Effect of Power-Law Fluid Behavior on Nusselt Number of a Circular Disk in the Forced Convection Regime

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Forced convection heat transfer is investigated from a thin disk in power-law fluids over wide range of conditions such as: Reynolds number, $1 \leq Re \leq 100$, Prandtl number, $1 \leq Pr \leq 100$, power-law index, $0.4 \leq n \leq 1.8$, and disk thickness to diameter ratio, $t/D = 0.01$, 0.025, 0.05, and 0.075. The wide range of values of the power-law index spanned here covers both shear-thinning as well as shear-thickening fluid behavior. These results also elucidate the influence of the type of thermal boundary conditions, i.e., constant wall temperature condition (CWT) and constant heat flux condition (CHF) prescribed on the disk surface. Extensive results are presented in terms of the local and average Nusselt numbers to delineate the effect of each of the influencing parameters, Re, Pr, n, t/D for each thermal boundary condition. Limited results are also included here at vanishingly small values of the Peclet number to understand the behavior in the creeping flow condition. Finally, the present numerical results on the average Nusselt number have been consolidated in the form of a predictive equation to facilitate the interpolation of the present data for intermediate values of the parameters and/or a priori estimation of the average Nusselt number in a new application. [DOI: 10.1115/1.4042784]

Keywords: disk, forced convection, Nusselt number, power-law index, Reynolds number, Prandtl number

1 Introduction

Due to frequent occurrence of power-law fluid behavior in a broad range of industrial settings, significant developments have occurred during the past 20 years or so on heat transfer from variously shaped bodies in such fluids in the forced convection regime. The currently available results for three-dimensional shapes like a sphere [1–4], spheroids [5–7], hemisphere [8], and for two-dimensional shapes like a circular cylinder, e.g., see Refs. [9] and [10], all seem to suggest that it is possible to realize up to 30–40% enhancement in heat transfer in shear-thinning fluids (power-law index smaller than unity) and, as expected, shear-thickening behavior ($n > 1$) has deleterious impact on heat transfer under appropriate conditions. Of course, the extent of augmentation or reduction in heat transfer is strongly dependent on the degree of advection (values of Reynolds number and Prandtl numbers) and on the shape of the object. It is thus possible to regulate the rate of cooling or heating by using suitable additives like clays or polymers in minute quantities to cooling water, as is routinely done in the thermal treatment of foodstuffs [11]. A thorough search of the literature reveals the paucity of results on heat transfer from circular disks even in Newtonian fluids, let alone in power-law fluids. Notwithstanding the fundamental significance of such model shapes, disk-like particles are encountered in electronic components, blood, food, and pharmaceutical product sectors and in ceramic-related industries. Thus, a systematic study of the forced convective heat transfer from a thin disk to power-law fluids serves as a precursor to understanding the rheological behavior of suspensions of disk-like particles, and their heating and cooling characteristics. Such suspensions also find applications as drilling muds in oil wells, ceramic and coating applications, etc. Hence, this work is concerned with the forced convection heat transfer from a disk to power-law fluids. However, it is instructive to begin with a terse discussion of the scant results currently available on heat transfer from a disk in Newtonian and non-Newtonian fluids.

Since much of the pertinent literature has been thoroughly reviewed in a recent work [12], only the salient points are recapitulated here. A reasonable body of knowledge is available on the fluid mechanical aspects of a disk in Newtonian, power-law and Bingham plastic fluids, e.g., see Ref. [12]. In a few studies, the hydrodynamic behavior of freely falling disks has been investigated, both experimentally and numerically, while others have studied the boundary effects, and the wake and drag characteristics for the transverse flow across a stationary disk. This body of information has been reviewed recently by Gupta et al. [12]. They also noted the scant information about the forced convection heat transfer from a disk in Newtonian fluids. For instance, Kobus and coworkers [13–15] have experimentally studied heat transfer in air from a disk in the forced, natural, and mixed convection regimes. For the case of forced convection regime, their Nusselt number results span the range of Reynolds number, $30 \leq Re \leq 3 \times 10^4$ and for disk thickness-to-disk diameter ratio varying from 0.062 to 0.2. Furthermore, they also reported the effect of the orientation of the disk to be negligible for $Re < 1000$. Similarly, the value of (t/D) did not influence the value of the average Nusselt number for the range of conditions spanned by their study. Limited numerical results on the forced convection heat transfer from a disk oriented normal to the direction of the flow of Newtonian and Bingham plastic fluids over the ranges of conditions as Reynolds number, $Re \leq 150$ and Prandtl number, $1 \leq Pr \leq 150$ have been reported in Ref. [12]. However, no comparisons were made between their numerical results and the experimental results of Kobus and Shumay [13]. To the best of our knowledge, no other results are available on heat transfer from a heated disk in Newtonian fluids and none at all in the power-law fluids. This work endeavors to fill this gap in the current literature.

In particular, extensive numerical results on heat transfer from a disk subjected to the two common thermal boundary conditions,

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i.e., constant wall temperature (CWT) and constant heat flux (CHF) are reported here for wide ranges of conditions as: $1 \le \mathrm{Re} \le 100$, $0.4 \le n \le 1.8$, $1 \le \mathrm{Pr} \le 100$ and disk thickness to diameter ratio $(t/D) = 0.01$, 0.025, 0.05, and 0.075. The heat transfer characteristics are studied here in terms of the local and average Nusselt numbers to elucidate the influence of each of the pertinent parameters. Indeed, preliminary comparisons between the present predictions and the scant experimental results [13] are encouraging. Also, limited results are presented here in the limit of zero Reynolds number and low Peclet number region.

2 Problem Formulation

Consider the incompressible flow of a power-law fluid (U_0, T_0) past a thin circular disk (of diameter D and thickness t), as shown in Fig. 1(*a*) and similar to that used in Ref. [12]. This is equivalent to the so-called impinging flow in the study of Kobus and Shumway [13]. The disk surface is maintained either at a constant heat flux condition, q_o (CHF condition), or at a constant temperature of T_w (CWT condition). The flow is assumed to be axisymmetric, steady, incompressible, and laminar over the range of conditions spanned here. In Newtonian fluids, the axisymmetric flow regime is encountered up to about Re = 100–130, and it is assumed here to be so far a power-law fluid also. In order to keep the level of complexity at a tractable level, the thermophysical properties (k, m, n, C, ρ) are considered constant over the small temperature difference ($\Delta T = T_w - T_0 = 5\mathrm{K}$) employed here and the viscous dissipation term is also neglected in the energy equation as the maximum value of the Brinkman number ($\mathrm{Br} \ll 0.01$) is rather small. Thus, there is no coupling between the velocity and temperature fields, and these can be obtained sequentially. Based on these simplifying assumptions, the dimensionless forms of the governing equations, namely, continuity, momentum, and energy are given as

$$\nabla \cdot \boldsymbol{U} = 0 \tag{1}$$

$$(\boldsymbol{U} \cdot \nabla)\boldsymbol{U} = -\nabla p + \frac{1}{\mathrm{Re}} \nabla \cdot \boldsymbol{\tau} \tag{2}$$

$$(\boldsymbol{U} \cdot \nabla)\xi = \frac{1}{\mathrm{Re\,Pr}} \nabla^2 \xi \tag{3}$$

For an incompressible fluid, the extra stress tensor, $\boldsymbol{\tau}$, is given as

$$\boldsymbol{\tau} = 2\eta\dot{\gamma} \tag{4a}$$

For a power-law fluid model, the viscosity (η) is expressed as

$$\eta = m(I_2/2)^{(n-1)/2} \tag{4b}$$

Here, m is the fluid consistency index, n is the power-law index, and I_2 is the second invariant of the rate of strain tensor [16].

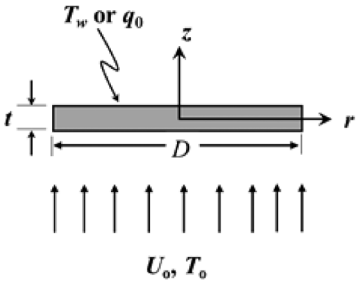
In this work, the truly unconfined flow condition is realized by situating the disk at the center of a cylindrical domain of diameter and length ($2L$), as shown in Fig. 1(*b*), along with the coordinates system used here. The relevant boundary conditions are given as

$$\bullet\ \text{At the inlet}\ (z = -50):\ U_r = 0, U_z = 1,\ \xi = 0 \tag{5}$$

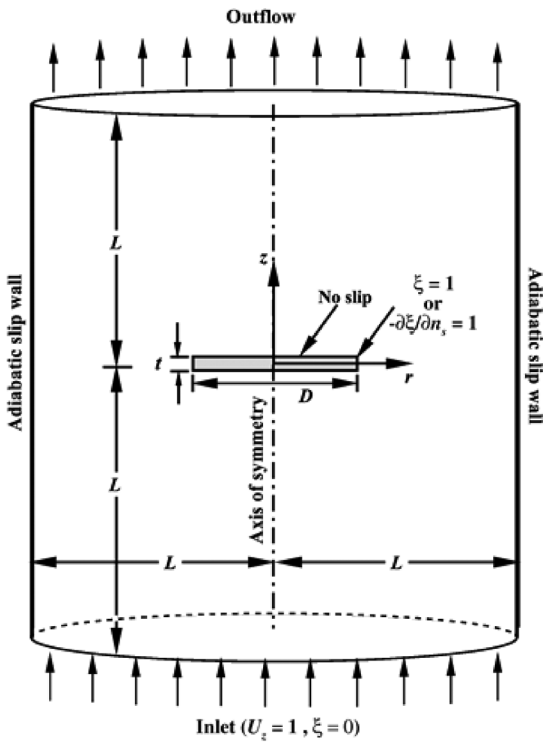
$$\bullet\ \text{At the outlet}\ (z = 50):\ \partial\phi/\partial z = 0\ \text{ where }\ (\phi = U_r,\ U_z \text{ or } \xi) \tag{6}$$

$$\bullet\ \text{On the surface of disk}:\ U_r = U_z = 0 \text{ and } \xi = 1\ (\text{CWT}) \text{ or } -\partial\xi/\partial \boldsymbol{n}_s = 1\ (\text{CHF}) \tag{7}$$

In order to ensure that the walls of the tubular domain do not influence the flow *per se*, these walls are treated as slip ($\partial U_z/\partial r = 0,\ U_r = 0$) and adiabatic. In the aforementioned governing equations and boundary conditions, the length, velocity, pressure, and the stress components have been rendered dimensionless using D, U_o, ρU_o^2, and $m(U_o/D)^n$, respectively. The temperature is scaled as $\xi = (T - T_0)/(T_w - T_0)$ for the CWT condition and $\xi = (T - T_0)/(q_0 D/k)$ for the CHF condition. Once the temperature field is known, the local Nusselt number, Nu_L, is obtained as



(*a*)



(*b*)

Fig. 1 Schematics of (*a*) flow and (*b*) computational domains

$$\mathrm{Nu}_{L(\mathrm{CHF})} = \frac{hD}{k} = \frac{1}{\xi}\bigg|_{\text{surface}} \quad \text{and} \quad \mathrm{Nu}_{L(\mathrm{CWT})} = \frac{hD}{k} = -\frac{\partial \xi}{\partial n_s}\bigg|_{\text{surface}}$$

and its surface averaged value, $\mathrm{Nu}_{\mathrm{avg}}$, is obtained by integrating the local values on the surface of the disk. Finally, the scaling arguments suggest the local and average Nusselt number in the present case to be influenced by four dimensionless numbers, namely, power-law index (n), Reynolds number ($\mathrm{Re} = \rho U_0^{2-n} D^n / m$), Prandtl number ($\mathrm{Pr} = (Cm/k)(U_0/D)^{n-1}$), and disk thickness to diameter ratio (t/D) for a given boundary condition prescribed on the surface of the disks. This work endeavors to explore and develop these functional relationships.

3 Numerical Solution Methodology and Choice of Computational Parameters

The nonlinear partial differential equations (Eqs. (1)–(3)) were numerically solved using the finite element-based solver COMSOL MULTIPHYSICS (version 4.3a) with a relative convergence criterion of 10^{-6}. Since a detailed description of the numerical solution methodology, choice of numerical parameters like type and density of computational mesh, type of solver, etc., is available in Ref. [12], only the salient features are described here. Based on an extensive exploration, suffice it to add here that the size of tubular computational domain of $L/D = 50$ was found to be adequate for the present results to be free from boundary effects for $\mathrm{Re} \geq 1$. Similarly, owing to the expected steep gradients close to the disk, this region was meshed using very fine triangular cells (extended up to distance of $6D$ in both axial and lateral directions) while non-uniform quadrilateral cells were used to map the remaining part of the domain. As noted earlier, since the flow is expected to be symmetric about $r = 0$, computations have been carried out only in half of the domain while using the symmetry boundary condition at the center plane.

The results reported in this work have been obtained with grids consisting of 83,942, 84,158, and 86,680 elements (half-domain) for $t/D = 0.01$, 0.05, and 0.075, respectively. This selection is based on the combination of conditions in terms of the extreme values of $\mathrm{Re} = 100$, $\mathrm{Pr} = 100$ and $n = 0.4$ and 1.8 when the boundary layer thickness will be minimum. Similarly, a limited full domain, time-dependent simulations were also performed, especially for the extreme values of Re, Pr, n, t/D, etc., to ascertain the validity of the occurrence of steady flow regime. Indeed, the flow and temperature fields are found to be steady over the range of conditions embraced in this work. Finally, since limited results were also obtained in the limit of $\mathrm{Re} \to 0$ and Peclet number of order 0.1, much longer domain ($L/D = 500$) was used for this purpose.

4 Results and Discussion

New extensive numerical results on the forced convection heat transfer from a thin circular disk subject to the thermal boundary conditions of the CWT and CHF are reported here for the following ranges of conditions: $1 \leq \mathrm{Re} \leq 100$, $1 \leq \mathrm{Pr} \leq 100$, $0.4 \leq n \leq 1.8$ and $t/D = 0.01$, 0.025, 0.05, and 0.075. Initially, the effects of the pertinent parameters Re, Pr, n and that of the thermal boundary conditions (CWT or CHF) on the heat transfer characteristics (local and average Nusselt number) are investigated for a fixed value of $t/D = 0.05$ and then further results are presented for additional three values of $t/D = 0.01$, 0.025, and 0.075 in order to delineate the role of this parameter on the average Nusselt number. However, before proceeding with a detailed discussion of the new results, some benchmark comparisons are carried out to ascertain the reliability and accuracy of the present results.

4.1 Validation of Results. As extensive validation with the available literature on the drag and average Nusselt number in Newtonian fluids are already presented in Gupta et al. [12], only additional comparisons are reported here. Table 1 shows a comparison of the present values of drag coefficient for $t/D = 0.025$ with those of Nitin and Chhabra [17] and an excellent agreement (within $\pm 1.8\%$) is seen to exist between the two results based on the finite volume method used in Ref. [17] and the finite element method in the present case. As there are no prior numerical results available on heat transfer from a thin disk in Newtonian fluids over this range of Reynolds numbers, the present results on the local Nusselt number from an oblate spheroid (aspect ratio = 0.5) in power-law fluids are compared with those of Srinivas and Ramesh [6], Fig. 2(*a*). The two results are in close agreement (within $\pm 3\%$). Similarly, Fig. 2(*b*) shows a comparison in terms of the average Nusselt number and the divergence between the two results never exceeded 3%. Finally, based on their results, Kobus and Shumway [13] put forward the following correlation for the average Nusselt number for a disk

$$\mathrm{Nu}_{\mathrm{avg}} = 2.22\ \mathrm{Pr}^{1/3}\ \mathrm{Re}^{0.34} \tag{8}$$

This equation is based on the results in a wind tunnel and for $30 \leq \mathrm{Re} < 1000$. Figure 3 shows a comparison between the present values and the predictions of Eq. (8) for the few data points in the common range. The experimental results are seen to be consistently higher by up to 20% than the present predictions. This is not at all surprising because the free convection effects are always, howsoever small, present in such experimental work. This will increase the Nusselt number above the pure convection value especially at low Reynolds numbers. Besides, the mean error associated with Eq. (8) is of the order of $\sim 10\%$. Bearing in mind these factors, the correspondence seen in Fig. 3 is regarded to be fair and acceptable. These comparisons coupled with the other validations reported in our previous study [12] inspire confidence in the reliability of the present numerical results.

4.2 Local Nusselt Number. The local Nusselt number on the surface of the disk indicates the temperature gradient normal to the surface for the CWT condition and the local temperature on the surface for the CHF condition. Figures 4 and 5 show representative results on the distribution of the local Nusselt number along the surface of disk. Owing to the symmetry about the z-axis, results are shown only for one-half of the disk. For both boundary conditions, the trends are qualitatively similar for Newtonian ($n = 1$), shear-thinning ($n = 0.4$) and shear-thickening ($n = 1.8$) fluids, except for the fact that the shear-thinning ($n < 1$) is seen to promote heat transfer whereas shear-thickening ($n > 1$) impedes it with reference to the corresponding values in Newtonian media under otherwise identical conditions. This effect becomes increasingly significant with the increasing values of the Reynolds number or Prandtl number or both. Conversely, at low Peclet numbers, e.g., $\mathrm{Re} = 1$, $\mathrm{Pr} = 1$, the power-law index exerts virtually no or very little influence on the local Nusselt number. This is so because in the conduction-dominated region, the viscosity of the heat transfer medium is largely irrelevant.

Now, turning our attention to the role of the thermal boundary condition prescribed on the surface of the disk, the two peaks seen in Fig. 4 correspond to the severe bending of the streamlines and isotherms at the two points B and C on the rim of the disk. Strictly

Table 1 Comparison of the total drag coefficient with that available in power-law fluids

	Nitin and Chhabra [17]			Present		
n	$\mathrm{Re} = 1$	$\mathrm{Re} = 20$	$\mathrm{Re} = 100$	$\mathrm{Re} = 1$	$\mathrm{Re} = 20$	$\mathrm{Re} = 100$
0.4	31.217	2.2035	1.0425	30.736	2.1759	1.0611
0.6	30.052	2.2576	1.0905	29.538	2.2757	1.1100
0.8	26.616	2.3867	1.1647	26.229	2.4219	1.1815
1.0	23.335	2.5755	1.2570	23.269	2.5722	1.2747

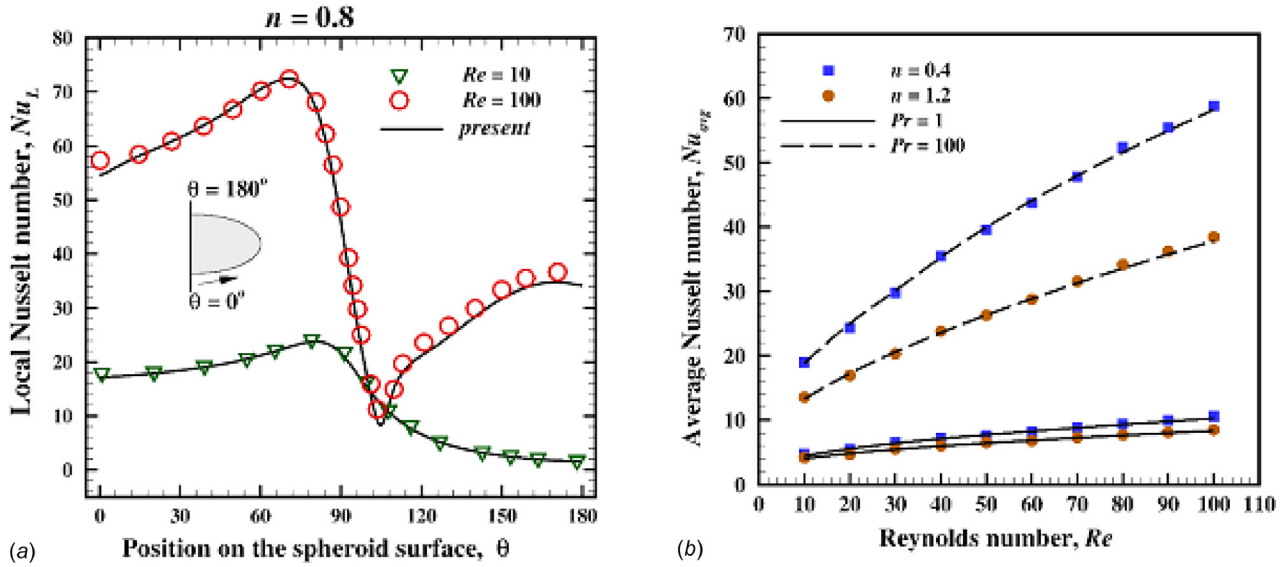


Fig. 2 Comparison of (*a*) the local Nusselt number distribution at Pr = 100 and (*b*) the average Nusselt number for an isothermal spheroid (aspect ratio = 0.5) with that of Srinivas and Ramesh [6]; lines: present results

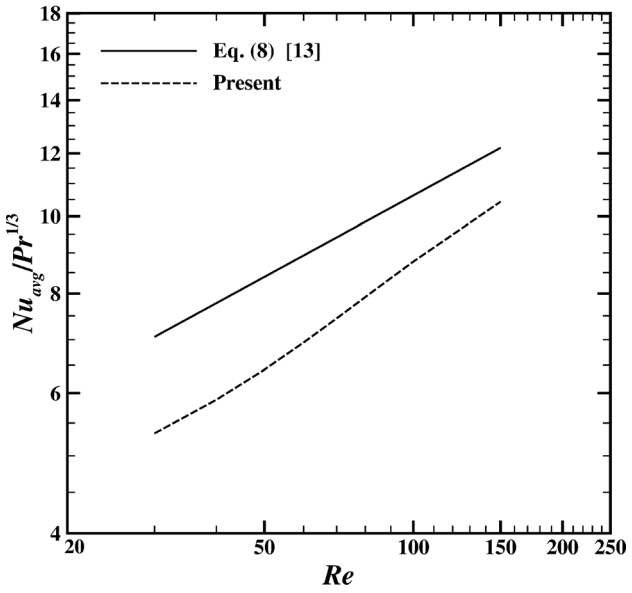


Fig. 3 Comparison of the present average Nusselt number for an isothermal disk in air with experimental results [13]

speaking, the temperature gradient (and hence the local Nusselt number) is indeterminate at these points. However, due to the use of a very fine mesh in this region, the variation is seen to be continuous at these locations in Fig. 4. On the other hand, such an uncertainty in evaluating the values of Nu_L at these singularities is of little importance because the corresponding heat transfer area is identically zero. In contrast, for the case of CHF condition (Fig. 5), only one peak is seen because in this case, the Nusselt number is given by the local temperature on the surface of the disk, which can be evaluated much more reliably than the temperature gradient.

Finally, it may be mentioned in passing that in both cases, the Nusselt number is seen to bear a positive dependence on the both Reynolds and Prandtl numbers irrespective of the value of the power-law index (n).

4.3 Average Nusselt Number. Although the local Nusselt number distributions provide useful insights into the detailed temperature fields and/or gradients, it is the surface-averaged value of the Nusselt number, which is often required in process design calculations. Dimensional considerations suggest the average Nusselt number to be determined by the values of t/D, Reynolds number, Prandtl number, and power-law index for a given boundary condition. This multivariable functional dependence is shown in Fig. 6 for both thermal boundary conditions. Based on a detailed examination of the results shown in Fig. 6 and of those not included here for the sake of brevity, the following observations can be made: the average Nusselt number bears the expected positive dependence on the Reynolds and Prandtl numbers due to the gradual thinning of the thermal boundary layers. As seen in Figs. 4 and 5, the dependence of the average Nusselt number on the power-law index is very weak at low Peclet numbers, e.g., $\mathrm{Pe} = 1$ due to poor advection under these conditions. Hence, the fluid viscosity is irrelevant here. However, with the increasing value of the Peclet number, advection increases, and the influence of the power-law index on heat transfer gradually rises. For instance, at $\mathrm{Pr} = 100$, $\mathrm{Re} = 100$, as the value of the power-law index varies from $n = 1$ to $n = 0.4$, the average Nusselt number increases by nearly $\sim 19\%$ and $\sim 28\%$ for the CHF and CWT conditions, respectively. Thus, shear-thinning behavior promotes heat transfer over and above that observed in Newtonian fluids. This can be ascribed to the lowering of the effective fluid viscosity and/or the thinner thermal boundary layers in shear thinning fluids than that in Newtonian media at fixed values of the Reynolds and Prandtl numbers. However, as the power-law index rises beyond $n > 1$, the fluid exhibits shear-thickening behavior and the effective fluid viscosity is higher compared to that in Newtonian fluids; as expected, this type of behavior lowers the heat transfer by up to about $\sim 17\%$ and $\sim 19\%$ for the CHF and CWT conditions, respectively. Furthermore, it can also be noted from Fig. 6 that the overall heat transfer is consistently higher in the constant wall temperature case than that in the constant heat flux case. Finally, it is worthwhile to consolidate the present numerical results by an expression of the following form:

$$\mathrm{Nu}_{\mathrm{avg}} = \mathrm{Nu}_0 + a\,\mathrm{Re}^{0.58}\,\mathrm{Pr}^{0.4}\left(\frac{3n+1}{4n}\right)^{0.81} \tag{9}$$

Where the best values of the Nu_0 and a for the two boundary conditions are

$$\mathrm{CWT}: \mathrm{Nu}_0 = 2.453, \quad a = 0.44$$

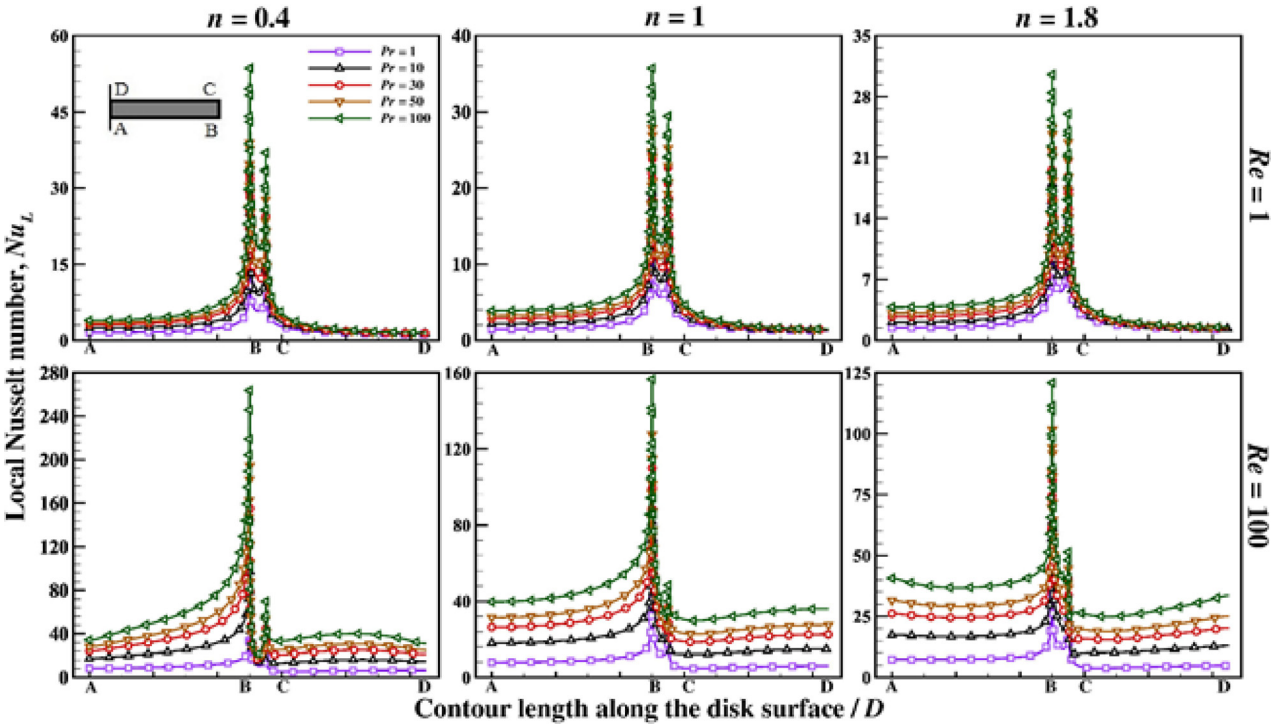


Fig. 4 Variation of the local Nusselt number along the surface of disk for CWT condition

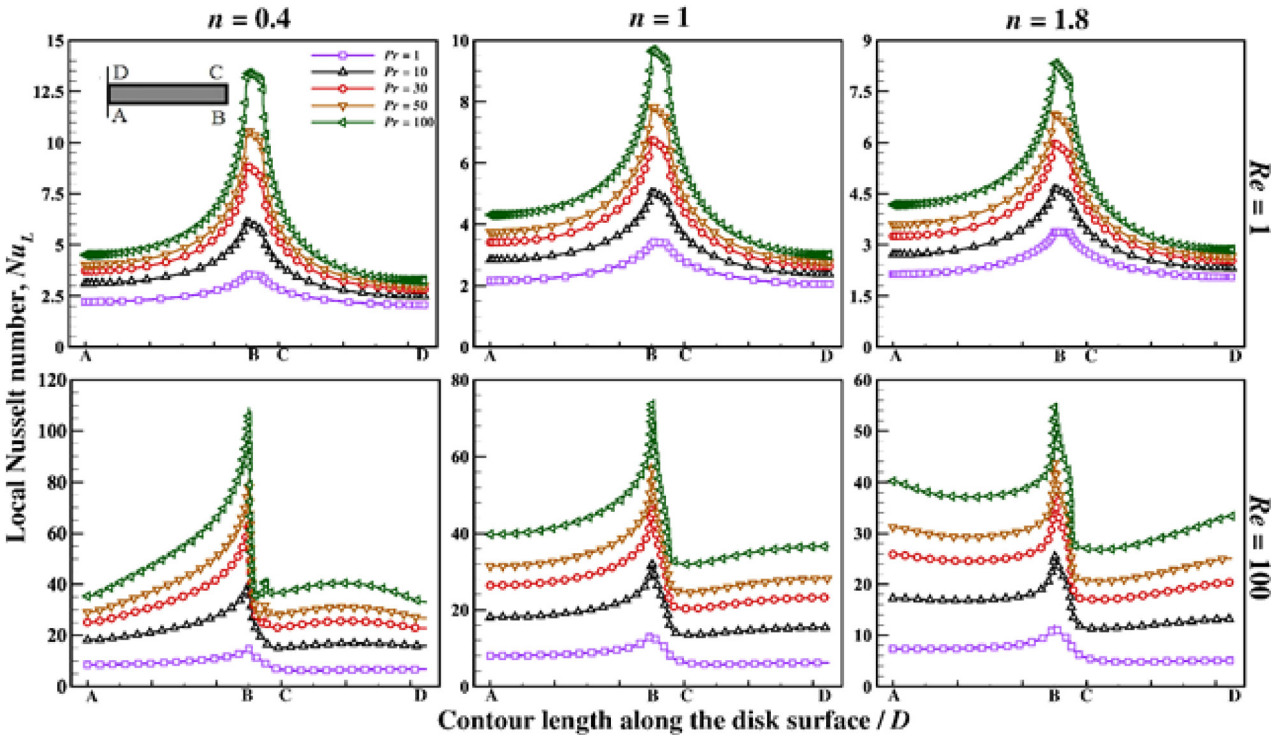


Fig. 5 Variation of the local Nusselt number along the surface of disk for CHF condition

$$\text{CHF: } \mathrm{Nu}_0 = 2.342, \quad a = 0.42$$

The average and maximum errors for Eq. (9) for both cases are ~5% and ~16% for 420 data points for each case. Suffice it to add here that only 26 data points (mostly in the transition range between the conduction- and convection- dominated heat transfer regimes) exhibit deviations greater than 10%, i.e., twice of the percentage mean error. Some remarks regarding the form of Eq. (9) and the values of the disposable parameters are in order. In the limit of $\mathrm{Re} \to 0$, Eq. (9) does reduce to the corresponding conduction limit (i.e., Nu_0) depending upon the type of boundary condition prescribed on the surface of the disk. The positive index of the term $((3n+1)/4n)$ captures the fact that the shear-thinning fluid behavior ($n < 1$) fosters heat transfer.

As noted earlier, limited results were also obtained for vanishingly small values of the Reynolds number (as small as 10^{-7}) and

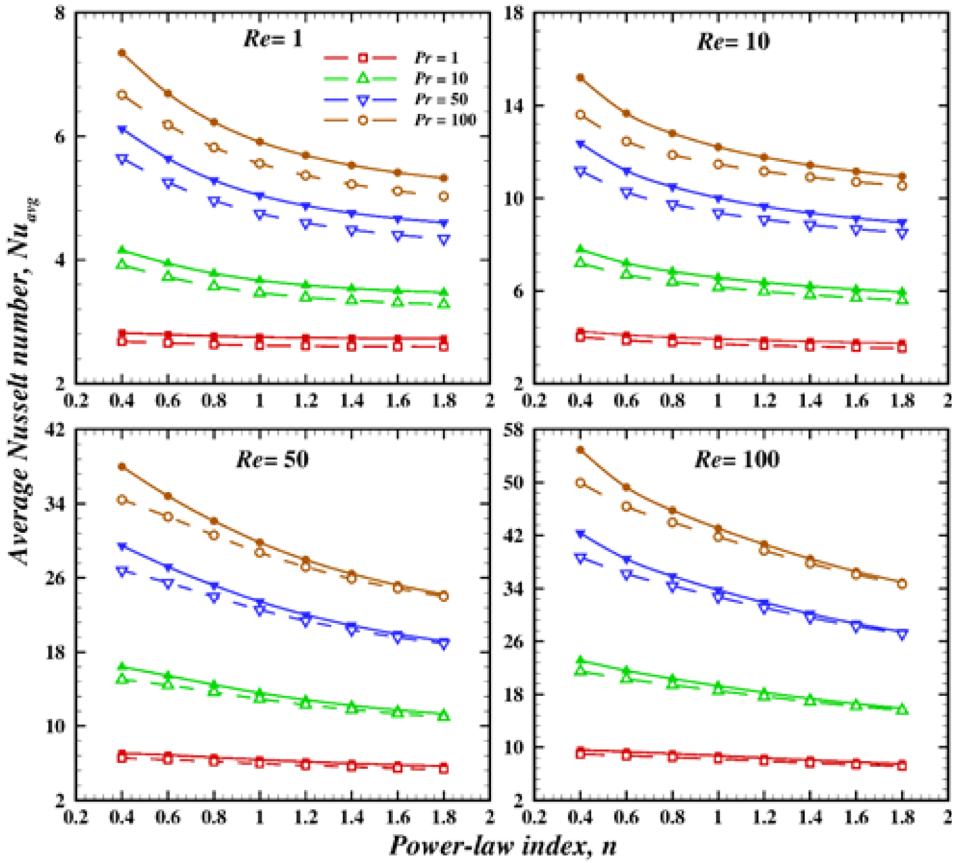


Fig. 6 Dependence of the average Nusselt number on the Reynolds number, Prandtl number, and power-law index (solid lines: Nu_{CWT}; dashed lines Nu_{CHF})

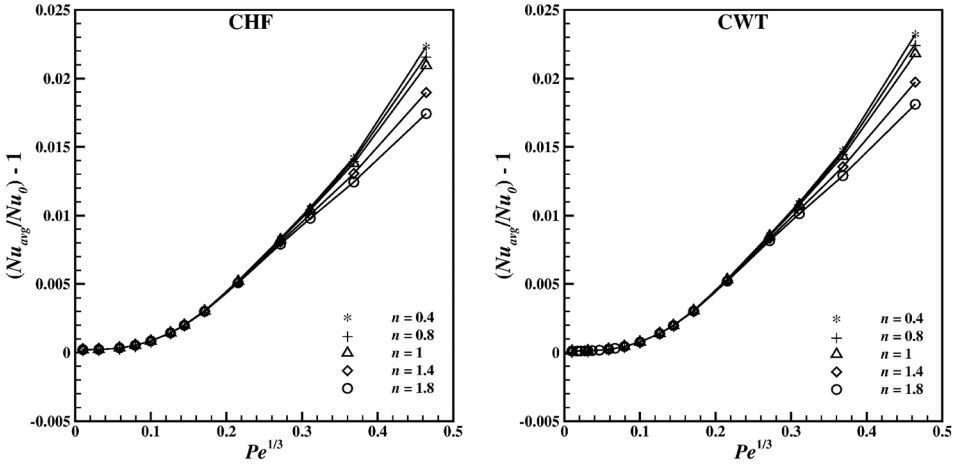


Fig. 7 Behavior of the average Nusselt number in the low Peclet number and Reynold number region

Peclet number (Pe $\lesssim$ 0.1) for $t/D = 0.05$. These results are shown in Fig. 7 for the two boundary conditions and for a range of values of the power-law index. First, as expected, the average Nusselt number approaches its conduction limit as Pe $\rightarrow$ 0. Second, the effect of the power-law index is seen to manifest itself at about $Pe^{1/3} \sim 0.3$, i.e., Pe $\sim$ 0.03. Beyond this value of Peclet number, $n < 1$ is seen to augment the value of Nusselt number and $n > 1$ reduces it with reference to the corresponding value in Newtonian fluids, i.e., $n = 1$. This behavior is qualitatively consistent with that observed for a sphere in Newtonian fluids.

4.4 Effect of Disk Thickness. Though the preceding discussion is restricted to the fixed value of the $t/D = 0.05$, the effect of disk thickness on the average Nusselt number is examined here by varying the value of $t/D = 0.01$, 0.025 and 0.075. The selection of these values is guided by the fact that as the value of t/D increases, a disk becomes a short cylinder and therefore, the velocity and temperature fields are expected to be three-dimensional due to the end effects and also heat transfer area increases. For instance, the contribution of the lateral surface area to the total heat transfer area increases from $\sim$2 to 13% as the value of t/D increases from

Table 2 Effect of disk thickness (t/D) on the average Nusselt number at Pr = 1 and 100 for a range of values of the power-law index and Reynolds number

		Nu_{CHF}			Nu_{CWT}		
Re	n	$t/D = 0.01$	$t/D = 0.05$	$t/D = 0.075$	$t/D = 0.01$	$t/D = 0.05$	$t/D = 0.075$
Pr = 1							
1	0.4	2.6905	2.6782	2.6633	2.8681	2.8168	2.8101
	0.8	2.6465	2.6306	2.6139	2.8205	2.7668	2.7586
	1.0	2.6317	2.6148	2.5976	2.8043	2.7500	2.7413
	1.2	2.6225	2.6049	2.5874	2.7942	2.7395	2.7306
	1.8	2.6110	2.5928	2.5751	2.7819	2.7269	2.7177
50	0.4	6.4203	6.5406	6.5909	7.0587	7.0627	7.0813
	0.8	6.0606	6.1661	6.2155	6.6351	6.6209	6.6395
	1.0	5.8391	5.9485	5.9996	6.3788	6.3709	6.3943
	1.2	5.6399	5.7547	5.8067	6.1494	6.1510	6.1785
	1.8	5.2069	5.3255	5.3748	5.6572	5.6740	5.7067
100	0.4	8.8702	8.9610	8.9599	9.6729	9.5905	9.5431
	0.8	8.3514	8.4691	8.4952	9.0915	9.0390	9.0132
	1.0	8.1030	8.1997	8.2227	8.7930	8.7228	8.6957
	1.2	7.8261	7.9043	7.9248	8.4673	8.3848	8.3590
	1.8	7.0185	7.1027	7.1338	7.5493	7.4959	7.4934
Pr = 100							
1	0.4	6.2111	6.4116	6.6743	6.8757	7.0028	7.2283
	0.8	5.5377	5.6637	5.8227	5.9362	6.0720	6.1998
	1.0	5.3255	5.4306	5.5610	5.6514	5.7997	5.9037
	1.2	5.1659	5.2572	5.3695	5.4459	5.6061	5.6954
	1.8	4.8710	4.9420	5.0293	5.0878	5.2724	5.3417
50	0.4	12.237	12.813	13.608	14.212	14.265	14.824
	0.8	11.083	11.407	11.868	12.234	12.311	12.635
	1.0	10.834	11.098	11.470	11.757	11.851	12.108
	1.2	10.617	10.845	11.161	11.400	11.489	11.706
	1.8	10.119	10.295	10.533	10.703	10.766	10.927
100	0.4	34.749	34.698	34.401	33.770	40.018	39.390
	0.8	30.543	30.593	30.614	30.412	33.324	32.999
	1.0	28.679	28.710	28.747	28.626	30.786	30.533
	1.2	27.081	27.125	27.187	27.115	28.692	28.518
	1.8	23.802	23.873	23.987	23.991	24.600	24.552

0.01 to 0.075. Table 2 shows representative results elucidating the effect of t/D on the average Nusselt number for both thermal boundary conditions. At Pr = 1, it is found that the disk thickness does not have much influence on heat transfer as the maximum increase in Nu_{CHF} and Nu_{CWT} is ~3% and ~2% only. The effect of t/D was seen to be entwined with the values of Re, Pr, and n. While the overall effect of (t/D) over the range of conditions spanned here was rather weak, the maximum change in the value of Nu_{avg} was of the order of ~10% for about Pe = 100 at the lowest values of power-law index $n = 0.4$ used here, which progressively dropped to ~5.5% at Pe = 10^4. The effect of t/D was also seen to diminish with the increasing value of power-law index and/or Peclet number. This is probably due to the two competing mechanisms. Increasing value of Pe leads to the thinning of thermal boundary layer; some of this gain is offset by the fact that, for fixed values of Re and Pr, thermal boundary layer thickness increases with the increasing value of power-law index. This interplay is further accentuated because the power-law index is an independent dimensionless parameter on its own, in addition to its occurrence in the definition of the Reynolds and Prandtl numbers. The weak influence of (t/D) on the average Nusselt number seen here is in line with the experimental results of Kobus and Shumway in air [13].

5 Conclusions

In this work, the forced convection heat transfer for a heated disk (isothermal or isoflux) in power-law fluids has been studied for the following ranges of conditions: $1 \le Re \le 100$, $1 \le Pr \le 100$, $0.4 \le n \le 1.8$ and $t/D = 0.01$, 0.025, 0.05, and 0.075. Broadly speaking, for a fixed value of t/D and for a given thermal boundary condition, shear-thinning fluid behavior ($n < 1$) augments heat transfer over and above that observed in Newtonian fluids. This is so due to the lowering of the effective fluid viscosity. Conversely, shear-thickening fluid behavior ($n > 1$) impedes heat transfer as compared to that in Newtonian fluids otherwise under identical conditions. The average Nusselt number is not very sensitive to disk thickness at low values of Peclet number; this effect is slightly accentuated at high values of Peclet numbers. The average Nusselt number exhibits the classical positive dependence on the Reynolds and Prandtl numbers. Also, the heat transfer is found to be somewhat higher for an isothermal disk than that for the constant heat flux condition. Finally, simple correlations for the average Nusselt number are reported here, which can be used to interpolate the present data for the intermediate values of the pertinent parameters.

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Nomenclature

Br = Brinkman number
$((m(U_o/D)^{n-1}U_o^2/k(T_w - T_o))$ for CWT,
$(m(U_o/D)^{n-1}U_o^2/q_oD)$ for CHF)
C = thermal heat capacity of fluid, J kg^{-1} K^{-1}
D = diameter of the disk, m
h = local heat transfer coefficient, W m^{-2} K^{-1}
I_2 = second invariant of the rate of strain tensor, s^{-2}
k = thermal conductivity of fluid, W m^{-1} K^{-1}

L = size of the computational domain, m
m = flow consistency index, Pa·s^n
n = power-law index
n_s = unit vector normal to the surface of disk
$\mathrm{Nu_{avg}}$ = average Nusselt number
$\mathrm{Nu_L}$ = local Nusselt number
$\mathrm{Nu_0}$ = Nusselt number at the conduction limit
Pe = Peclet number (Re × Pr)
Pr = Prandtl number $(Cm/k(U_0/D)^{n-1})$
q_o = Constant surface heat flux on the surface of the disk, W m^{-2}
Re = Reynolds number $(\rho U_0^{2-n} D^n/m)$
t = disk thickness, m
T = temperature of fluid, K
T_o = temperature of the fluid in free stream, K
T_w = constant temperature on the surface of the disk, K
U_r, U_z = velocity components in r- and z- directions
U_o = inlet velocity, m s^{-1}
U = velocity vector
r, z = cylindrical coordinates, m

Greek Symbols

$\dot{\gamma}$ = rate of strain tensor
η = scalar viscosity function, Pa s
ξ = temperature of fluid $(T - T_o/q_o(D/k))$ for CHF
$(T - T_o/T_w - T_o)$ for CWT
ρ = density of fluid, kg m^{-3}
τ = extra stress tensor
ϕ = dependent variable
∇ = del operator

Subscripts

o = refers to free stream condition
w = refers to disk surface condition

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